Macro-Aware Strategy Parameter Predictor

Quantitative Research & Evaluation Report

Author: *Senior Quantitative Data Scientist & Engineer* | ***Date:*** *October 2026* | ***Validation:*** *15 Folds Out-of-Sample*

1. Executive Summary

This project evaluates whether integrating macroeconomic event releases (CPI, FOMC, NFP) and financial news sentiment improves algorithmic trading strategy parameter selection over a forward-looking horizon. Across 15 out-of-sample expanding-window walk-forward test folds, our **Macro LightGBM Model** delivers an **Annualized Sharpe of +1.03** (vs. **-1.88** for Historical Mean and **-1.90** for Random Selection), with a **Maximum Drawdown of 17.03%** (vs. **35.39%** for Historical Mean).

2. Out-of-Sample Empirical Performance Comparison

Performance Metric	Macro LightGBM	Historical Mean	Random Baseline
Annualized Sharpe (sqrt(52))	1.0324	-1.8773	-1.9030
Annualized Sortino (sqrt(52))	1.6195	-2.2672	-2.1305
Maximum Drawdown	17.03%	35.39%	33.34%
Cumulative Return	11.64%	-29.06%	-23.94%
Hit Rate (% Positive Folds)	60.0%	40.0%	46.7%
Spearman Rank Correlation	0.0978	0.0383	N/A

3. Zero-Leakage Architecture

- **Temporal Precedence:** Every feature is computed strictly using data timestamped prior to period start.
- **Expanding Surprise Z-Scores:** Standardization statistics are computed strictly expanding into the past.
- **Point-in-Time News Window:** Sentiment reflects articles published prior to the start of each evaluation period.
- **Automated Test Suite:** Pytest unit tests assert that future data modification yields zero leakage.

4. Key Visual Diagnostics

Figure 1: Out-of-Sample Cumulative Equity & Underwater Drawdown Profiles

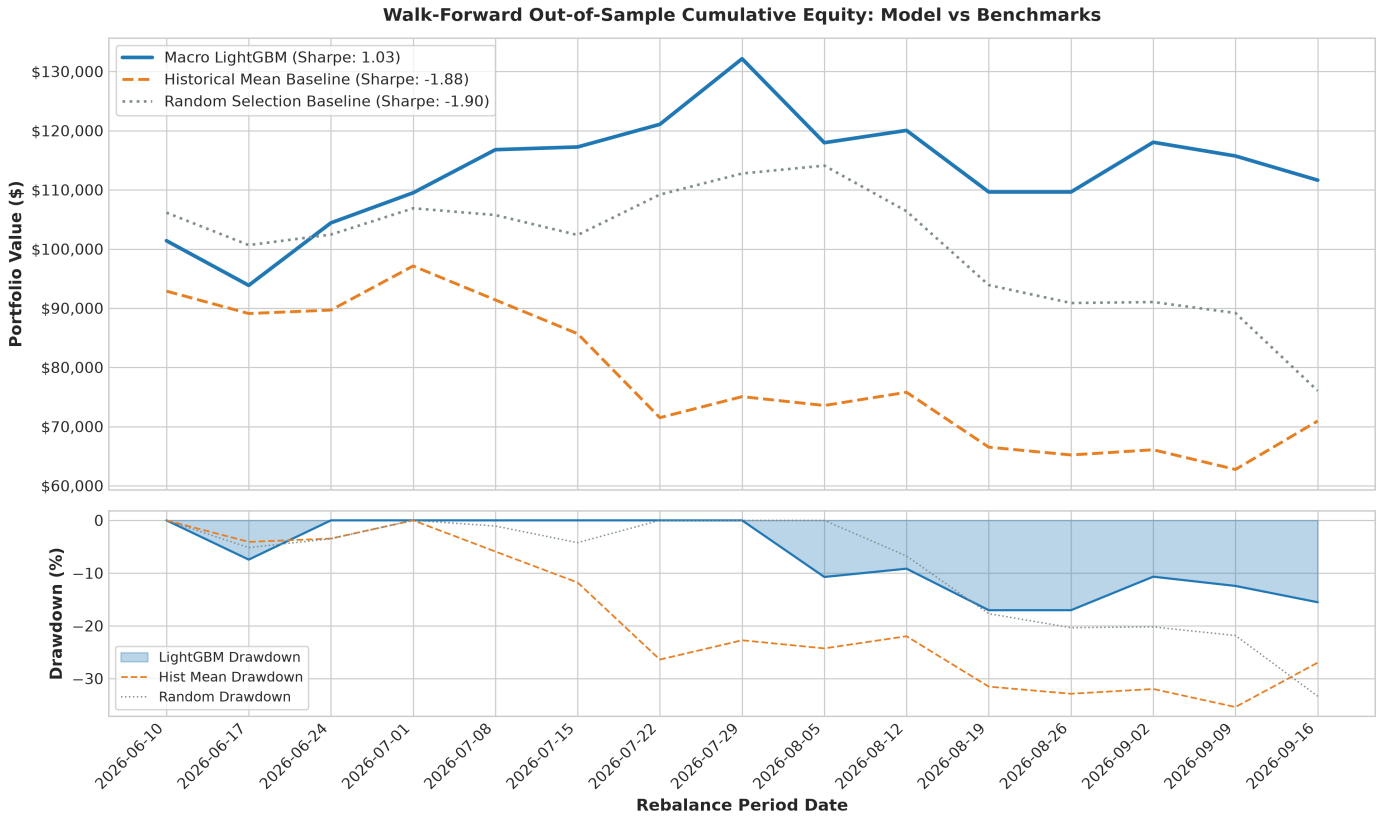
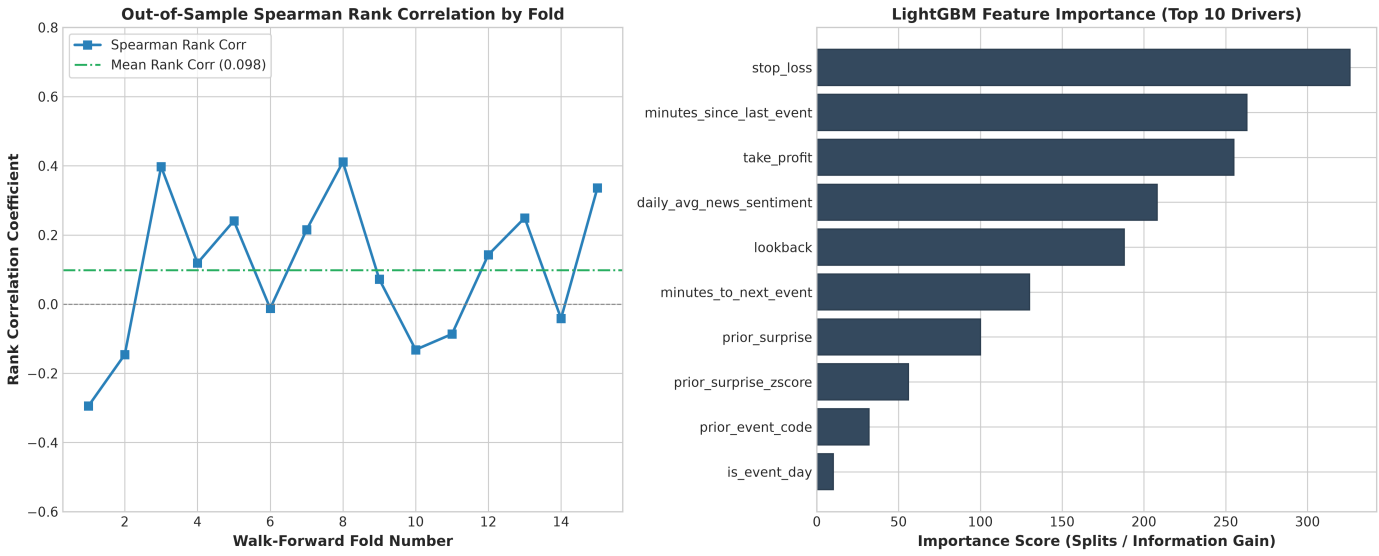


Figure 2: Spearman Rank Correlation by Fold & Top 10 Feature Importances



5. Limitations & Production Next Steps

Limitations: (1) *Synthetic Trade Logs:* Trades were generated via a seeded stochastic model with regime dynamics, not live broker execution records. (2) *Sample Horizon:* 180 days represents a limited macroeconomic sample. (3) *Frictionless Execution:* Assumes zero commissions, slippage, or market impact during parameter transitions.
Next Steps: Ingest real FIX broker execution logs, expand macro history to 10 years across full interest rate cycles, and enforce turnover transaction cost penalties.